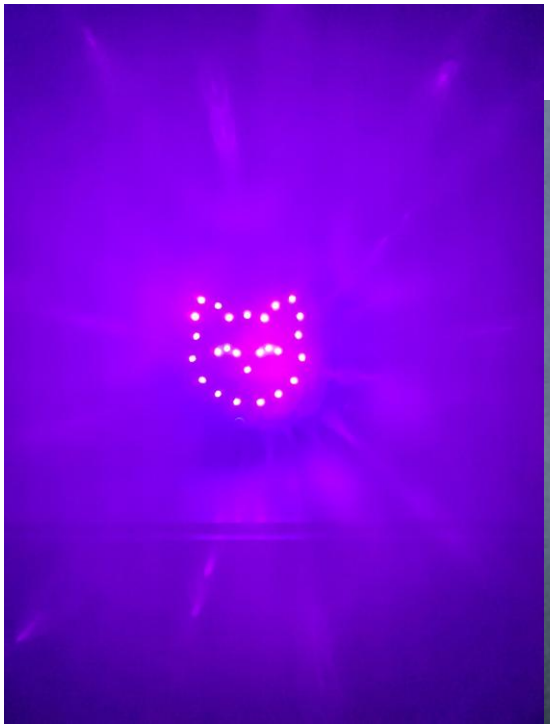
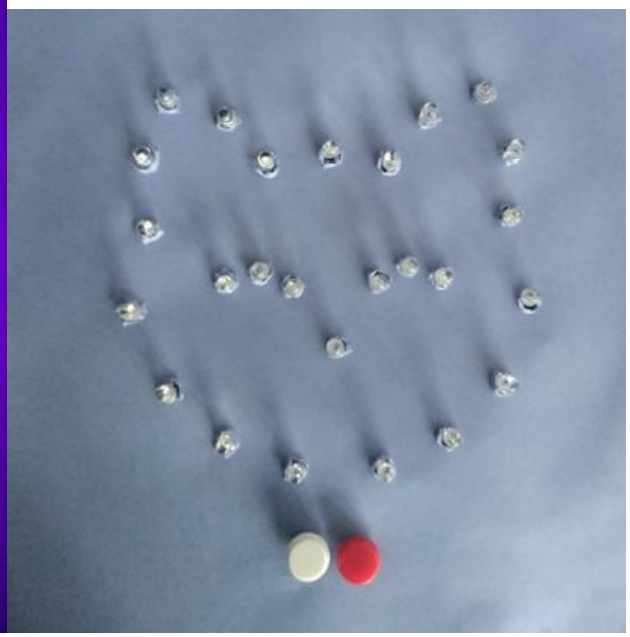
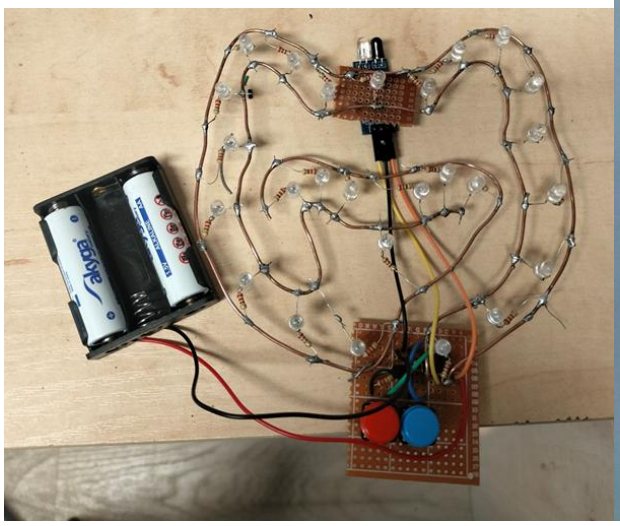
PROJECT

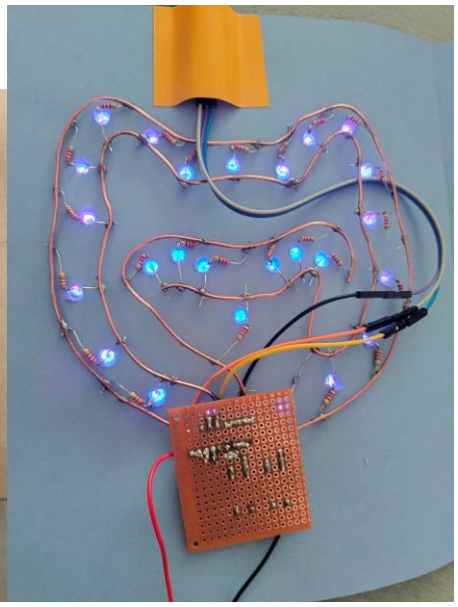
“Touch-Controlled LED Driver — Firmware for ATmega/ATtiny85 with PWM Animation and Brightness Control”

PROJECT PHOTOS









HOW IT WORKS

A touch-controlled LED lighting driver — upon detecting a gesture, it triggers a smooth LED pulsing animation with gamma correction. Brightness is adjusted in real time using two physical buttons.

The entire circuit is embedded inside a sheet of paper, as visible in photos.

> The key technical feature I wanted to implement in this project is an advanced LED animation technique. The curve used in the firmware to override the PWM modulation values is a modified sinusoid raised to an appropriate power. This adjustment accounts for the nonlinear perception of brightness by the human eye (gamma correction).

Using a time-dependent function requires that the animation always starts from a specific phase of the sinusoid (phase = 0.5) — otherwise the whole animation would appear chaotic and unpredictable.

It is worth noting that the phase variable increments by a fixed step every 1 ms, which ensures a repeatable and predictable animation regardless of the processor load.

Full source code and schematics available on GitHub: https://github.com/kojscezary-commits/led_Project